

A Rapid Compression Machine Investigation of Oxidation and Auto-Ignition of *n*-Heptane: Measurements and Modeling

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n-Heptane oxidation and auto-ignition in a rapid compression machine is studied in the low and intermediate temperature regimes at high pressures. Experimental ignition delay times and some phenomenological aspects related to knock in engines are presented, providing additional information at lower temperatures on previously published delays from shock tube experiments. The products of oxidation are identified and time profiles are measured during a two-stage ignition process. Eight C₇ heterocycles, heptenes, lower 1-alkenes, aldehydes, and carbon monoxide are the main species. Their origin is discussed in relation to the isomerization and decomposition of heptylperoxy radicals. The high selectivity observed in the formation of lower 1-alkenes is explained by the scission of the β C—C bond of the β -hydroperoxyheptyl radicals weakened by the presence of oxygen atoms. Numerical simulation of the experiments with Warnatz's comprehensive chemical mechanism gives satisfactory results for cool flame and total ignition delays, but fails to reproduce the detailed chemistry before auto-ignition.

INTRODUCTION

The oxidation and auto-ignition of *n*-heptane has been extensively studied in the last decades, as this hydrocarbon is one of the primary reference fuels used for the determination of the RON or MON index of gasoline. Short ignition delays have been measured at high temperatures and pressures in shock tubes experiments [1–4]. Species concentration profiles of initial, intermediate, and final products have been obtained from low to high temperature regimes using jet-stirred flow reactors [5–8]. Specific experiments have been conducted in CFR engines [9–10] and fundamental features observed in a rapid compression machine have been recently reported for *n*-heptane autoignition [11]. Along with experimental studies, detailed mechanisms, including both low and high temperature chemistry, have been developed [12–15]. These mechanisms are very complex and include important steps for which reliable kinetic data are not always well-known. They have to be validated using experimental results, before being used for predictions over a large range of conditions.

The purpose of this work is to provide new data on oxidation and auto-ignition of *n*-heptane in engine-like conditions using a rapid compression machine and to compare the experimental results with the predictions gener-

ated by an extensive mechanism. Both ignition delay times and chemical species' histories before auto-ignition have been obtained in the intermediate range of temperature.

EXPERIMENTAL

Rapid Compression Experiments

The oxidation and auto-ignition experiments were conducted in an original, right-angle, original, rapid, compression machine. A complete description of this apparatus has been published elsewhere [16–17] and only the main characteristics of the machine will be recalled. The stroke of the compressing piston is 199 mm and the compression ratio $\rho = 9.8$. The combustion chamber is a cylinder of 5 cm diameter and volume 37.9 cm³ equipped with windows and ports. Before compression the gas mixtures were introduced into the preheated compressing cylinder at a known temperature and pressure and compressed in one stroke without rebound. Light emission and pressure were recorded during and after compression until auto-ignition occurred. The ignition delays were measured on the recorded pressure time profiles from the end of compression to the first and second jumps of the ignition phenomena. As the pressure trace did not exactly follow the adiabatic law, the temperature of

the compressed gas, T_c , was calculated for an adiabatic core gas from the initial pressure, P_0 , and temperature T_0 , the compressed pressure P_1 , and the ratio of specific heats ($\gamma = C_p/C_v$) by:

$$\int_{T_0}^{T_c} \frac{\gamma}{\gamma - 1} \frac{dT}{T} = \ln \frac{P_1}{P_0},$$

taking into account the variation of γ with temperature (NASA tables). A numerical simulation of gas motion in a rapid compression machine has suggested that the hot core gas generated at the end of compression is virtually adiabatic and spans approximately 85% of the diameter of the chamber [20]. In situ measurements of gas temperature are in good agreement with the adiabatic core gas hypothesis [21].

Measurements were performed at different compressed charge densities ($P_0 \rho / RT_0$). For each charge density the temperature of the mean compressed gas was varied between 600 and 900 K by changing the nature and proportions of inert gases (CO_2 , N_2 , Ar). Engine-like conditions for dilution and equivalence ratio ($\phi = 1$) were selected. The possible influence of the time of compression (20–60 ms) was tested. In another series of experiments, the chemical evolution of the gas mixture during the delay was interrupted by piercing a thin diaphragm inserted in a port and quenching the reaction by adiabatic cooling of the mixture into a collecting vessel. The products were analysed by GC and GC/MS. A detailed description of the rapid sampling device has been given previously [18].

Analysis of Products

The identification of the products was made by mass spectrometry. The constituents with less than seven carbon atoms were separated on a 25 m Poraplot Q capillary column and the C_7 constituents on a 50 m 5%-diphenyl-95%-dimethylpolysiloxane capillary column. The quantitative analysis was made with another chromatograph fitted with three capillary columns (25 m Poraplot Q, 25 m dimethylpolysiloxane, 25 m 5 Å molecular sieve). An internal standard of 2% neon was introduced

into the reacting mixture. Two analyses of the mixture were made for each compression experiment. First the analysis of C_1 to C_6 compounds was made with the Poraplot Q column. The permanent gases were quickly eluted and directed to the molecular sieve for further separation and detection by thermal conductivity. The constituents eluting later were detected by flame ionization. The same procedure was applied with the dimethylpolysiloxane column, especially for the analysis of C_7 constituents. The detectors were calibrated for each species or group of species.

Modeling

An extensive mechanism developed by Warnatz [19] was adapted to our own format in order to model the evolution of the reacting mixture after compression. The adaptation brought out some deficiencies of the model, because all the coefficients of thermodynamical data of the NASA format (related to heat capacities and standard enthalpies of ketohydroperoxides) were found to equal zero. More realistic values were calculated according to the additivity method proposed by Benson and incorporated into the mechanism. Ignition delays and the evolution of chemical species were calculated with this adapted version of the mechanism for a homogeneous adiabatic system at top dead center (TDC). All calculations were performed with Chemkin and Senkin programs from Sandia.

RESULTS AND DISCUSSION

Ignition Delays

Typical time profiles of pressure change and light emission during and after compression of the reactive mixture are presented in Fig. 1 for a two-stage auto-ignition process. The first stage of ignition, which occurs after an induction period, t_1 , measured from TDC, is characterized by a pressure increase of about 1 bar and a pulse of light. After this cool flame event the hot ignition leads to a sharp and important increase of pressure and light emission. A total delay time t measured from TDC is observed for this phenomenon. The auto-ignition pro-

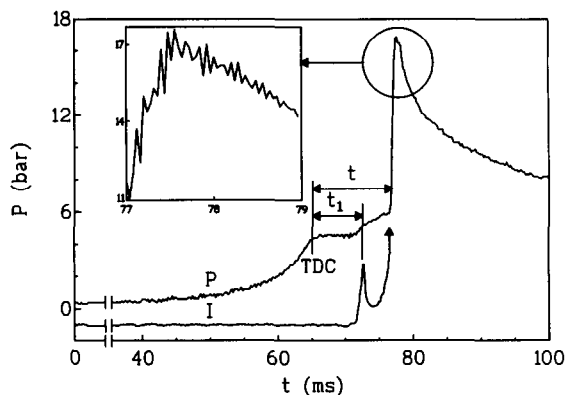


Fig. 1. Pressure change (P) and light emission (I) during the compression and the two-stage auto-ignition of a stoichiometric n -heptane-"air" mixture. $\rho = 9.8$, $T_0 = 355$ K, $P_0 = 219$ torr.

cess is associated with oscillations of the pressure after the onset of ignition, as can be seen in Fig. 1. They present a strong intensity and more regularity at higher initial pressures. These severe oscillations may be related to the occurrence of knock in engines and used to define a knock intensity factor [22], which is particularly high for n -heptane (RON = 0).

The influence of pressure and temperature on the ignition delays of n -heptane is presented in Fig. 2. For the lower initial pressures ($P_0 = 130$ and 162 torr) the variation of the total delay with core gas temperature shows clearly the usual inverse dependence of the

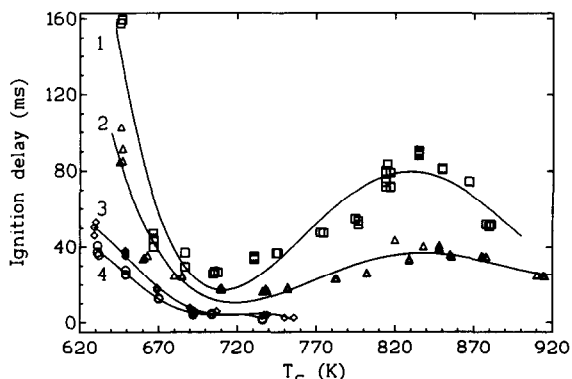


Fig. 2. Auto-ignition delays of stoichiometric n -heptane-"air" mixtures versus core gas temperature T_c . The symbols represent the experimental points and the curves the mean experimental values. The compression ratio $\rho = 9.8$ and $T_0 = 355$ K. 1. $P_0 = 130$ torr. 2. $P_0 = 162$ torr. 3. $P_0 = 313$ torr. 4. $P_0 = 406$ torr.

reaction rate of hydrocarbons against temperature. This is the region of negative temperature coefficient for the rate. As the temperature increases, there is a progressive weakening of the pressure and light signals associated with the cool flame and auto-ignition changes from a two-stage to a one-stage process. At higher initial pressures ($P_0 = 313$ and 406 torr) the ignition delays are very short and, as a result, accurate measurements could not be obtained at temperatures above 760 K. The higher reactivity of n -heptane than n -butane [18] is characterized by shorter auto-ignition delays and by a lower temperature limit under which ignition is no longer possible.

Evidence of the first stage reaction of the very reactive n -heptane/air mixture having occurred during the final stages of compression have been reported elsewhere [11]. Therefore special attention was devoted to this particular point. Different series of experiments were performed at low charge density ($P_0 = 130$ torr) and two times of compression (34 and 60 ms). This was facilitated by the particular features of our right angle RCM, which allows one to change the pressure in the air gun very easily. The results are reported in Fig. 3 for both total and cool flame delays. No important changes of the delays of the cool flame, which

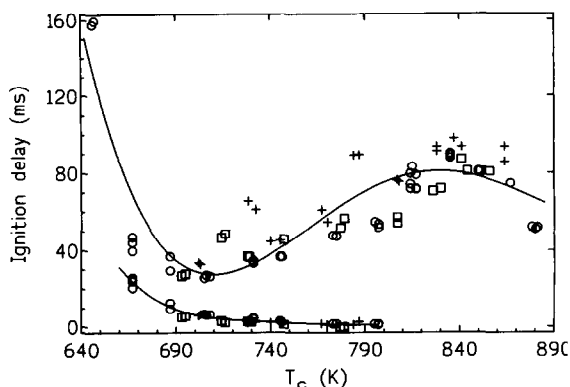


Fig. 3. Influence of time of compression (t_c) on the cool flame delay (lower line) and total delay (upper line) of n -heptane-"air" mixtures. The symbols represent the experimental points and the curves the mean experimental values. $\rho = 9.8$, $T_0 = 355$ K, $\phi = 1$, $P_0 = 130$ torr. $t_c = 34$ ms, (+); $t_c = 60$ ms (first set of measurements ($\circ$) and second set after modifications ($\square$), see text)

is the first well-marked event occurring after compression, are observed. In these experimental conditions the delay of the cool flame does not depend on the rate of compression; also the amplitude of the cool flame, which characterizes the low temperature reactivity, drastically decreases with temperature. The conditions for a negative temperature coefficient are the same for both times of compression. The scatter of the experimental results for the total delay times may be attributed more to the precision of measurements carried out after a two years interval after disassembling and ressembling the machine for modifications, than to a conjectural reactivity during the final stages of compression. Moreover, no chemiluminescence was observed during compression and, if undetected reactions occurred during compression, they did not decrease the delays more than any uncontrolled fluctuation. From these experiments it became evident that pure *n*-heptane did not react to a perceptible extent during compression, because in this study the selected initial pressure and the resulting compressed gas pressure were not too high. Similar conclusions will be drawn from a chemical analysis of the products.

Recently, ignition delays for *n*-heptane have been measured at higher pressures and temperatures in shock tube experiments [4]. Measurements in a shock tube and a RCM have been compared for stoichiometric *n*-heptane/air mixtures. In Fig. 4 the global ignition delay times (t) are plotted logarithmically versus the inverse of the temperatures calculated behind the reflected wave in the shock tube or in the core gas at TDC in the RCM. So, ignition delay times from 10^{+2} to 10^{-2} ms can be compared over large ranges of temperature 650–1300 K) and pressure (3 to 40 bar). From this extensive set of results it appears that the *n*-heptane/air mixtures show an Arrhenius dependence for the ignition delay, both at low and high temperatures and also the same particular feature of a negative temperature coefficient for the rate at any pressure in the intermediate range of temperature. It is noteworthy that an increase in pressure by about a factor of ten produces an increase by about 100 K of the negative temperature zone.

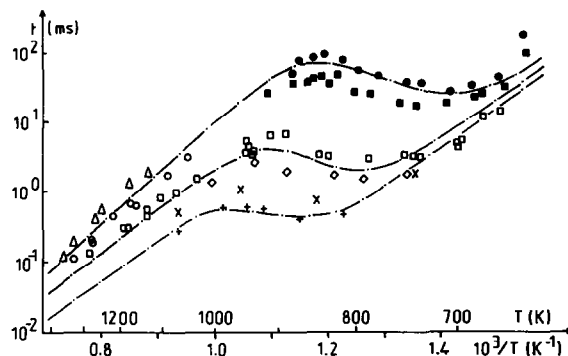


Fig. 4. Total ignition delay times of stoichiometric *n*-heptane–“air” mixtures over an extended range of pressure and temperature. +, x, $\diamond$, $\square$, $\circ$, Δ are results of Ciezki and Adomeit obtained in shock tube experiments [4] at respectively 42, 30, 19.3, 13.5, 6.5 and 3.2 bar. $\bullet$, $\blacksquare$ are results obtained in RCM experiments [this work] for initial pressure P_0 of respectively 130 and 162 torr (pressure at TDC varying from about 2.7 to 4.5 bar). The dashed-dotted lines are the results of computation of Warnatz for 3, 13.6 and 40 bar [4].

Nature and Time Evolution of the Products before Autoignition

Chemical analysis of the mixture was performed for a relatively low charge density ($P_0 = 162$ torr) to identify and follow the chemical evolution of the products formed during the oxidation process leading to auto-ignition. Experiments were achieved for different core gas temperatures and special attention was focused on the evolution of products during a two-stage auto-ignition.

Products' Formation

Figure 5 shows two representative chromatograms obtained from the total GC/MS ion current of a mixture removed at a sampling time between the cool and normal flame. The lighter constituents (1 to 25) were well separated on the poraplot Q column (chromatogram a) and the C_7 constituents on the siloxane column (chromatogram b). The identification of the heptenes and lighter species was readily achieved by comparing their mass spectra with the spectra of a reference library and their retention times with the retention times of pure standard substances. In contrast, the C_7 oxygenated constituents 26 to 33 were more difficult to recognize as no other C_7 -O-heterocycles than oxiranes 31 and 32 were

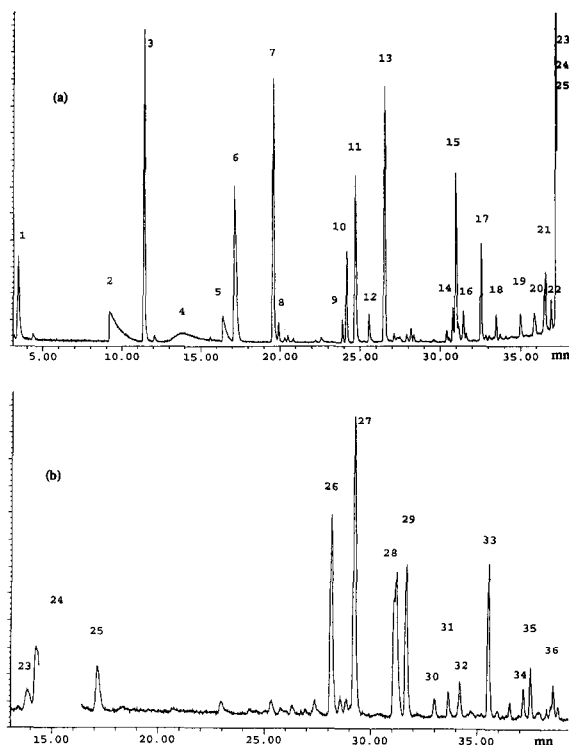


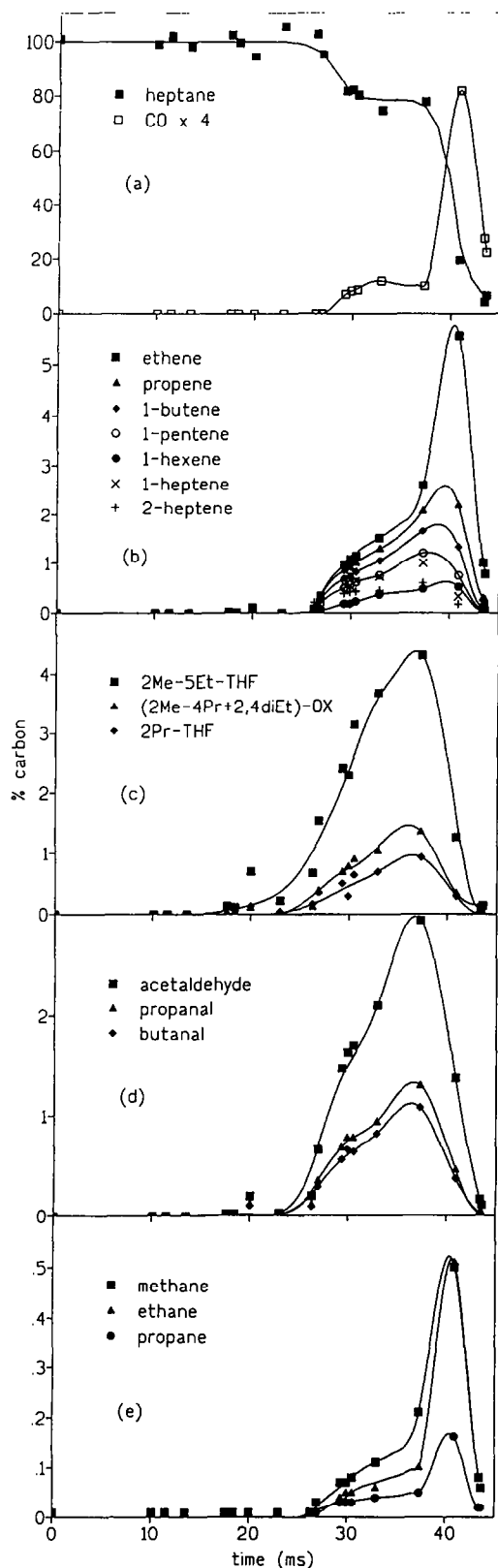
Fig. 5. Typical GC/MS chromatograms of the products observed after the appearance of the cool flame with (a) a Poraplot Q column and (b) a siloxane column. 1. Ethylene, 2. Water, 3. Propylene, 4. Formaldehyde, 5. Methanol, 6. Ethanal, 7. 1-Butene, 8. 1,4-Butadiene, 9. unidentified, 10. Propenal, 11. Propanal, 12. Acetone, 13. 1-Pentene, 14. unidentified, 15. Butanal, 16. Butanone, 17. 1-Hexene, 18. 2-Butenal, 19. 2-Methyl-3-ethyloxirane, 20. unidentified, 21. Pentanal, 22. 2-Methylhexane (in heptane), 23. 1-Heptene, 24. Heptane and 3-Heptene unresolved, 25. 2-Heptene, 26. 2-Methyl-5-ethyltetrahydrofuran, 27. 2-Methyl-5-ethyltetrahydrofuran, 28. 2,4-Diethyloxetane and 2-Methyl-4-propyl oxetan unresolved, 29. 2,4-Diethyloxetane and 2-Methyl-4-propyl oxetan unresolved, 30. 2-Ethyltetrahydropyran, 31. 2-Ethyl-3-propyloxirane, 32. 2-Methyl-3-butyloxirane, 33. 2-Propyltetrahydrofuran, 34. 3-Heptanone, 35. 2-Heptanone, 36. Pentyloxirane and heptanal unresolved.

found in reference works and pure products were not easily available. Thus, the mass spectra of the unknown chromatographic peaks were thoroughly analyzed with the help of fragmentation paths published for similar compound of lower molecular masses. Peaks 26 and 27 presented nearly identical mass spectra with very little difference in only one ion fragment intensity. They are fully compatible with the fragmentation paths for the geometric iso-

mers of 2-methyl-5-ethyltetrahydrofuran. Peaks 30 and 33 were identified by the same procedure. Peaks 28 and 29 also displayed similar spectra but were found to be mixtures: selected ion chromatograms based on fragmentation paths for C_7 oxetanes supported the presence of two pairs of geometric isomers of 2,4-diethyloxetane and 2-methyl-4-propyloxetane eluting too closely to be resolved in the total ion chromatogram of Fig. 5b. Products 26, 27, 31, 32, 33, and 2-methyl-4-propyloxetane were previously found in static or flow experiments, as well as in a fired engine [23] and products 26 and 27 in a high pressure jet-stirred reactor [8], all fed with a mixture of *n*-heptane and air.

Concentration Profiles of the Products

To characterize the main reactions occurring before the negative temperature coefficient region, experiments were performed at $T_c = 667$ K and $P_c = 3.4$ bar. Under these conditions a two-stage auto-ignition was observed. The cool flame delay time was about 30 ms, whereas the second stage of ignition was observed for a total delay time of 41 ms after TDC. As shown in Fig. 6 (a-e), no important chemical reactions occurred during compression, nor during the induction period preceding the cool flame. *n*-Heptane consumption and product formation exhibit a more or less parallel and typical two-stage evolution with time. An important increase in the build-up of products is connected with the cool flame event, which consumes about 20% of *n*-heptane. After the cool flame all measured products increase again and finally go through a maximum as the second stage of ignition is taking place. 1-Heptene and 2-heptene are of the same order of magnitude. By contrast a high selectivity is observed in the formation of the lower alkenes: 1-hexene, 1-pentene, 1-butene. Significant amounts of C_7 -oxygenated heterocycles are associated with the cool flame, especially 2-methyl-5-ethyltetrahydrofuran (cis and trans), oxetanes, and 2-propyltetrahydrofuran. The concentrations of aldehydes have the same order of magnitude as C_7 -oxygenated heterocycles. They appear in the cool flame and disappear during the second stage of ignition in a way parallel to the C_7 heterocycles. Saturated hydrocarbons, such as



methane, ethane, and propane, may be considered as minor products.

The Chemical Mechanism

The chemical mechanism has to take into account every detected product. For the large range of temperature occurring in RCM experiments, both low- and high-temperature mechanisms must be considered. These mechanisms have been extensively described in the past and reviewed [24, 25]. Only a brief outline of the main reactions of peroxy radicals will be discussed here.

n-Heptyl radicals are the primary products of H-abstraction in the initiation step and in the main propagating reactions. Four different *n*-heptyl radicals denoted as iR (with $i = 1$ to 4) can be distinguished. They lead to the four corresponding peroxy radicals iRO_2 by reaction with oxygen. The behavior of peroxy radicals is of central importance to the formation of products in the low and intermediate ranges of temperature. By external H-abstraction from *n*-heptane they give branching via hydroperoxides decomposition. By internal H-atom abstraction they form hydroperoxyalkyl radicals iQOOH :



These isomerization reactions involve mainly H-transfer from primary (p) or secondary (s) C-atoms in the α , β , γ , and δ positions to the peroxidized C-atom. They are labeled as (1,4 p or s), (1,5 p or s), (1,6 p or s), and (1,7 p or s) isomerizations.

Two main decomposition reactions of the isomerized alkylhydroperoxy radicals are generally considered: (i) the breaking of the O—O bond in every iQOOH , giving O-heterocycles and iOH , (ii) the scission of the C—O bond of the α^iQOOH to give HO_2^i and the three possible *n*-heptenes:

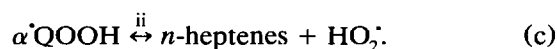
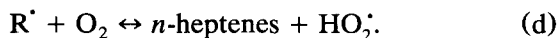
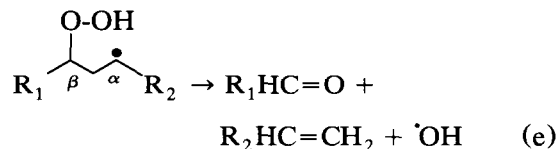


Fig. 6. (a-e). Experimental concentration profiles of the products in a two-stage auto-ignition of a stoichiometric *n*-heptane—"air" mixture. $P_{TDC} = 3.4$ bar, $T_c = 667$ K. Heptenes $\times 2$; all (c) $\times 2.5$.

The conjugate alkenes may also be produced by a direct reaction of oxygen on *n*-heptyl radicals or via the formation of a transient energized alkylperoxy radical [26]



Although the scission of C—C bonds in $\dot{Q}OOH$ is generally considered less probable than the scission of O—O bonds, special attention has to be paid to the hydroperoxyalkyl radicals formed by the (1, 5) isomerization of peroxy radicals. In this case the β C—C bond is probably weakened by the presence of an O atom on the β carbon. The breaking of the C—C bond leads to an aldehyde, a 1-alkene, and an $\dot{O}H$ radical, according to the following scheme:



($R_1 = H$ to C_4H_9 , $R_2 = C_4H_9$ to H)

This scheme presents an alternative path for the decomposition of β -hydroperoxyalkyl radicals to the simple O—O scission leading to oxetane.

Another point to consider is the reaction of the hydroperoxyalkyl radicals with oxygen, followed by the isomerization of the resulting species $\dot{O}OOOOH$ and the subsequent branching reactions, giving some complex compounds and two $\dot{O}H$ radicals:



In the past, dihydroperoxides have been isolated in the oxidation of *n*-heptane [27] and dicarbonyl products in the oxidation of 2-methylpentane [28]. More recently, evidence has been produced for the formation of complex species such as heptylketohydroperoxide, a branching agent responsible for autoignition in a CFR engine supplied with an air-*n*-heptane mixture [9].

The (1,4 p or s) to (1,7 p or s) isomerizations of *n*-heptylperoxy radicals (1RO_2 to 4RO_2) and the subsequent decomposition of the corresponding hydroperoxyheptyl radicals can lead to ten possible C_7 -oxygenated heterocycles: three oxiranes, three oxetanes, two tetrahydrofurans, and two tetrahydropyrans. All these heterocycles are listed in Table 1 together with the isomerization type of the corresponding peroxy radicals. As described in the preceding section, eight out of the ten heterocycles have been effectively identified in the products of the cool flame. Tetrahydrofurans and oxetanes are the main C_7 -oxygenated heterocycles formed in our experimental conditions.

Only 1-heptene and 2-heptene formed by reactions (c) and (d) were detected, because the chromatographic device does not separate 3-heptene from *n*-heptane. On the other hand, the observed selectivity in the formation of alkenes lighter than C_7 (1-hexene, 1-pentene, 1-butene) is an important experimental argument for the effective part taken by reactions (e) in the chemical mechanism at low temperatures.

The activation energies of the isomerization reactions of the peroxy radicals are still under discussion [24, 29, 30]. By contrast there is general agreement that O-heterocycle forma-

TABLE 1
 C_7 -Oxygenated Heterocycles^a

Oxirane	Oxetane	Tetrahydrofuran	Tetrahydropyran
2-pentyloxirane <u>[$^1RO_2(1,4 s)$; $^2RO_2(1,4 p)$]</u>	2-butyloxetane <u>[$^1RO_2(1,5 s)$; $^3RO_2(1,5 p)$]</u>	2-propyltetrahydrofuran <u>[$^1RO_2(1,6 s)$; $^4RO_2(1,6 p)$]</u>	2-ethyltetrahydropyran <u>[$^1RO_2(1,7 s)$; $^3RO_2(1,7 p)$]</u>
2-methyl-3-butyloxirane <u>[$^2RO_2(1,4 s)$; $^3RO_2(1,4 s)$]</u>	2-methyl-4-propyloxetane <u>[$^2RO_2(1,5 s)$; $^4RO_2(1,5 s)$]</u>	2-methyl-5-ethyltetrahydrofuran <u>[$^2RO_2(1,6 s)$; $^3RO_2(1,6 s)$]</u>	2,6-dimethyltetrahydropyran <u>[$^2RO_2(1,7 s)$]</u>
2-ethyl-3-propyloxirane <u>[$^3RO_2(1,4 s)$; $^4RO_2(1,4 s)$]</u>	2,4 diethyloxetane <u>[$^3RO_2(1,5 s)$]</u>		

^a The identified products are underlined

tion involves non-reversible reactions with activation energies of 14, 13, 3 and 0 kcal/mol for oxirane, oxetane, tetrahydrofuran and tetrahydropyran [24], respectively. The mechanism used here incorporates all the above reactions of peroxy radicals except the (1, 7) isomerization scheme and the reactions of hydroperoxyalkyl radicals through breaking of the β C—C bond. The isomers of heptenes, hexenes, and pentenes are not differentiated. Alkyl decomposition through β scission and isomerization of alkyl radicals via H atom abstraction are also taken into account. Finally, it includes 168 species and 904 elementary reactions.

Modeling of Ignition Delay Times and Concentration Profiles

Recently the shock tube ignition delays measured by Ciezki and Adomeit [4] for stoichiometric *n*-heptane air mixtures have been compared with the results predicted by Warnatz's extensive mechanisms [4]. Handwritten (about 900 reactions and 170 species) and automatically generated chemical mechanisms (about 2400 reactions and 620 species) give similar results [13, 15]. The total ignition delay times from numerical simulations [4] are reproduced in Fig. 4 as dash-dotted lines for three pressures (3, 13.6, and 40 bar). They are in good agreement with the experimental data obtained in shock tubes. Our own results for lower temperatures and pressures are of the order of magnitude predicted by modeling. However our experimental results have been obtained at constant mean charge density, instead of constant pressure. So new calculations were undertaken in our conditions for modeling cool flame and total ignition delay times.

The experimental and calculated cool flame and total ignition delay times for an initial pressure P_0 of 162 Torr are presented in Fig. 7. The numerical simulation was performed for a homogeneous adiabatic system at calculated temperatures ($T_c \approx 640 - 940$ K) and measured pressures ($P_1 \approx 3 - 4.5$ bar) at TDC. It is clear that the adapted mechanism of Warnatz accounts for the evolution of both total and cool flame delay times with temperature. Experimental and calculated delay times are the same order of magnitude. The moderate

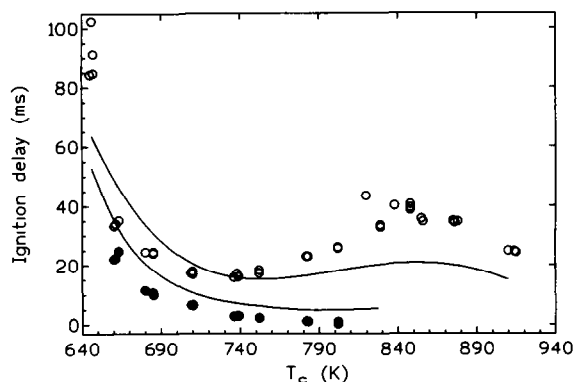
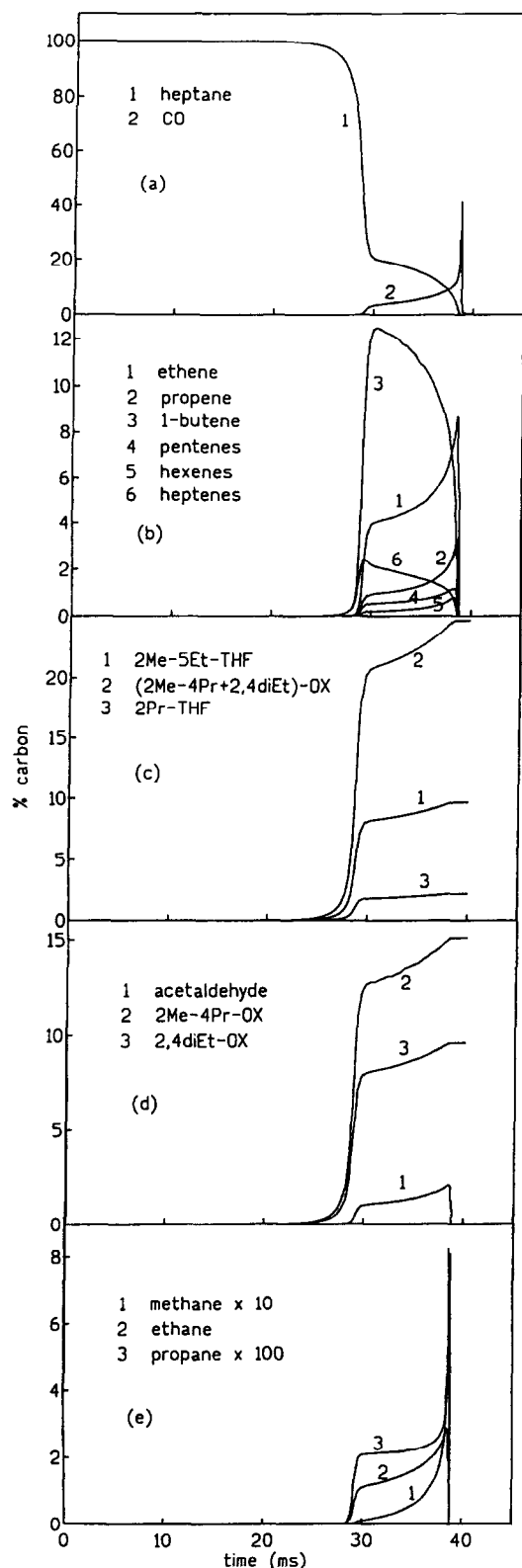


Fig. 7. Comparison between experimental and calculated ignition delay times of stoichiometric *n*-heptane—"air" mixtures at different core gas temperature T_c . $\rho = 9.8$, $T_0 = 355$ K, $P_0 = 162$ torr. Symbols are experimental measurements of cool flame ● and total ○ ignition delay times. Full lines are the calculated values with the detailed mechanism.

disagreement between experiments and modeling may be attributed to inaccuracies in the physical and chemical features of the model adopted to simulate the experiments.

The chemical evolution of the products at low temperature ($T_c = 667$ K, $P_{TDC} = 3.4$ bar) predicted by the model is shown in Fig. 8. The two-stage time profiles typical of the low temperature regime can be seen for *n*-heptane consumption and product formation, as in the experiments (Fig. 6). The modeled heterocycles, by contrast to the other products, do not vanish after ignition because they are treated as final products in the detailed mechanism.

There are other disagreements between the model and the measurements which should also be addressed. In the first place, the model predicts a much more significant consumption of *n*-heptane (about 80%) than observed (about 20%). This divergence hinders a direct comparison between predictions and measurements of the absolute quantities of products. To overcome the difficulty, the relative distribution of the carbon atoms in the products (after the cool flame) has been calculated for *n*-heptane consumption. The results are presented in Table 2. The proportion of carbon monoxide is underestimated by the model. Nevertheless, the experimental results are in good agreement with the general observation that carbon



monoxide is a major compound formed in a cool flame. The alkenes found experimentally do not verify the prediction, which favors 1-butene. The C_7 -oxygenated heterocycles are overestimated. A higher proportion of oxetanes than 2-methyl-5-ethyltetrahydrofurans is predicted. This prediction is clearly inconsistent with the experimental measurements, but can be explained in terms of the model, which overlooks a possible $\beta C-C$ bond breaking in the hydroperoxyalkyl radicals. This neglected pathway competes with oxetane formation and, as already mentioned, may also account for the experimentally observed selectivity in the formation of the C_6 to C_4 1-alkenes. Aldehydes, which appear significantly in the cool flame, are also predicted incorrectly. Finally, ketene, for which a high yield is predicted, has not been detected.

CONCLUSION

The autoxidation of *n*-heptane preceding second stage of ignition in the intermediate range of temperature and at high pressures presents a number of particular features relevant to the octane number of gasoline and to formation of pollutants in engines. In this study, the behaviour of *n*-heptane/air mixtures has been investigated after rapid compression and special attention has been given to auto-ignition delays, formation of molecular intermediates and numerical simulation of the results. *n*-Heptane exhibits a high reactivity characterized by a relatively short ignition delay and by a relatively low ignition limit, in accordance with the octane number of this fuel. For the selected experimental conditions, ignition occurs either in two stages with an intermediate cool flame or in one stage when the temperature is above 800 K. Strong pressure oscillations related to knock in the engine have been observed. The delay times show a remarkable negative dependence upon gas temperature in the range $T_c = 700$ –860 K. By comparing the results obtained in this work with published

Fig. 8. (a–e) Calculated concentration profiles of the products in a two-stage auto-ignition of a stoichiometric *n*-heptane–“air” mixture. Initial conditions: $P_{TDC} = 3.4$ bar, $T_c = 667$ K.

TABLE 2

Comparison Between Experiments and Modeling of the Relative Distribution of the Products After the cool flame for *n*-heptane consumption. $\Phi = 1$, $P_{TDC} = 3.4$ bar, $T_c = 667$ K, $t = 33$ ms

Products	% C Experimental	% C Calculated
Normalized consumption of <i>n</i> -heptane	100	100
Carbon Monoxide	11.79	5.84
1-heptene	1.46	n.c.
2-heptene	0.89	n.c.
3-heptene	m.	n.c.
all heptenes		2.16
1-hexene	1.47	n.c.
all hexenes		0.31
1-pentene	3.03	n.c.
all pentenes		0.75
1-butene	4.18	14.06
2-butene	0.2*	0.06
propylene	5.14	1.35
ethylene	6.02	5.35
2-pentyloxirane	t.	0.81
2-methyl-3-butyloxirane	0.6*	2.33
2-ethyl-3-propyloxirane	0.5*	1.89
2-butyloxetane	n.d.	4.38
2-methyl-4-propyloxetane	n.r.	16.10
2,4-diethyloxetane	n.r.	10.19
the two last oxetanes	1.64	26.29
2-propyltetrahydrofuran	1.10	2.29
2-methyl-5-ethyltetrahydrofuran	5.86	10.34
2-ethyltetrahydropyran	0.4*	n.c.
2,6-dimethyltetrahydropyran	n.d.	n.c.
formaldehyde	3*	5.29
acetaldehyde	8.37	1.45
propanal	3.75	n.c.
butanal	3.27	n.c.
ketene	n.d.	10.19
methane	0.44	0.04
ethane	0.24	1.65
propane	0.16	0.02
butane	0.1*	1.15

* estimated

n.r.: not resolved

n.d.: not detected

m.: masked by *n*-heptane

n.c.: not considered

t.: trace

data obtained in a shock tube, it appears that the particular region of the negative dependence occurs in the intermediate range of temperature at any pressure between 3 and 40 bar and that a ten-fold increase in pressure produces a shift of this region by about 100 K towards higher temperatures.

The detailed chemical analysis of the species' histories during a two-stage ignition has shown

a moderate consumption of initial *n*-heptane, an unselective formation of heptenes, a high selective formation of lower 1-alkenes, a transient appearance of a series of C_7 -oxygenated heterocycles and aldehydes, and the formation of lower saturated hydrocarbons and carbon monoxide. These results can be interpreted by the chemistry of the peroxy radicals, which result from initial H-abstraction and oxygen

addition. After an isomerization reaction, heterocycles and heptenes are formed simultaneously. Eight out of ten possible heterocycles have been identified. The selectivity in lower 1-alkene formation is supposed to be due to the scission of a C—C bond, weakened by the presence of an oxygen atom on the β carbon of the isomerized peroxy radicals.

The extensive mechanism of *n*-heptane oxidation developed by Warnatz, which is able to reproduce quite satisfactorily the ignition delay times measured in shock tubes, has been adapted and tested against our experimental results. Its validity has been confirmed as far as ignition delays and negative temperature coefficients are concerned, including the prediction of the cool flame delays. However, it is unable to predict correctly the history of *n*-heptane and its oxidation products before auto-ignition. This statement raises the important question of the pertinence of large detailed mechanisms for the prediction of overall phenomena, if they do not produce the details of chemistry in a realistic way. Global mathematical models inspired by simplified chemistry with adapted parameters may be more convenient in this case. On the other hand, detailed mechanisms need to be seriously improved before being used as a tool for the prediction of pollutants.

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